RCS Day 1

Residential Systems & Shop Safety

CTS Cohort 6 | Monday, July 20, 2026

Today's rule:
A professional does not guess. A professional checks, protects, and verifies.



One building. Many systems. One standard: verify before covering.

Today's flow

Start safe, learn the system, then build and test a simple plumbing assembly.

1	10:00	**Arrival + safety reset**
2	10:15	**RCS overview**
3	10:40	**PPE + shop rules**
4	11:10	**Power tool intro**
5	11:35	**Quiz + review**
6	12:30	**Rolling sink lab**
7	2:35	**Document + cleanup**
8	2:50	**Exit ticket**

Quiet quiz first. Hands-on work after the safety frame is clear.

Learning targets

By dismissal, students should be able to do these without guessing.

Name the major residential building systems.
Select PPE for common shop tasks.
Explain stop-work authority and tool accountability.
Identify common power tools and their main hazards.
Assemble a simple sink drain and P-trap.
Leak test, correct, and document the assembly.

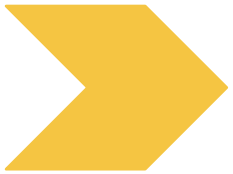
Today is about professional habits before capstone work starts.

What RCS means in CTS

Residential Construction Specialty is where technical habits become jobsite habits.

Work habits

Measure before cutting
Inspect before covering
Communicate before moving material
Clean before calling a task finished



Trade content

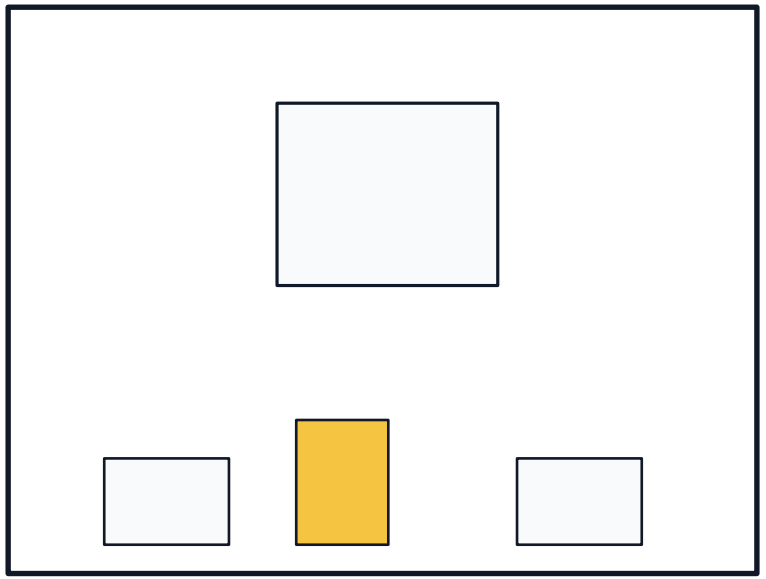
Framing
Electrical rough-in
Plumbing basics
Drywall and inspection

The habit is the portable skill.

The final wall project starts today

Every day points toward the individual 4 ft x 3 ft wall build.

Frame a small wall section.
Build a window rough opening.
Install two receptacle boxes.
Install one switch and light circuit.
Drywall the whole board with proper cutouts.
Explain and inspect the finished work.



4 ft x 3 ft individual wall

Residential systems map

Buildings are not random parts. They are connected systems.

Structure

Electrical

Plumbing

HVAC

Envelope

Finish

A good builder understands what will be hidden, protected, or affected by the next trade.

Professional expectations

Employers notice habits before they trust skill.

Show up ready to work.
Ask before touching unknown tools or systems.
Keep your area organized enough to be safe.
Tell the truth about mistakes quickly.
Document work so another person can understand it.

Today is about professional habits before capstone work starts.

Safety is not a speech

Safety is the way the work gets done.

PPE

Eyes, ears, hands, lungs, and clothing match the task.

Position

Stable feet, clear hands, clear blade path, no body parts in the line of fire.

Communication

Call out before cuts, moving material, energizing, or testing.

PPE for today

Use the protection that matches the hazard. More is not always better; wrong PPE is still wrong.

Eye protection

Required during cutting, drilling, fastening, and leak-test cleanup.

Gloves

Useful for carrying material. Not for rotating blades/bits.

Hearing protection

Required around saws and loud tools.

Dust control

Drywall and cutting dust need cleanup and breathing awareness.

Stop-work authority

Stopping is a skill. Guessing is not a method.

Something feels unsafe or unclear.
A tool is damaged or behaving oddly.
A cut, route, or connection does not match the plan.
Someone is in the path of material, blade, cord, water, or tool movement.
You cannot explain the next step.

Stop. Ask. Verify.

Tool accountability

A missing tool is not a mystery. It is a future problem.

1. **Check out**
2. **Inspect**
3. **Use correctly**
4. **Return clean**
5. **Report damage**

Same tool. Same place. Every time.

Power tool safety ladder

Students earn tool use through demonstrated control, not confidence.

1. **Identify tool**
2. **Inspect condition**
3. **Set material support**
4. **Check body position**
5. **Operate slowly**
6. **Shut down safely**
7. **Clean and return**

Confidence is not clearance. Demonstration is clearance.

Drill and driver safety

Small tools can still injure people and ruin work.

- Correct bit or driver tip
- Battery removed during bit changes
- Clamp or secure material
- Two hands when needed
- Low speed for starting
- Stop before stripping or splitting

Slow is smooth. Smooth is safe.

Saw safety introduction

A saw cut is a planned movement, not a surprise event.

Know the blade path
Support both sides of material
Keep hands outside the danger zone
Wait for full stop before moving off cut
Never bypass guards
Ask before any unfamiliar cut

Slow is smooth. Smooth is safe.

Measurement habit

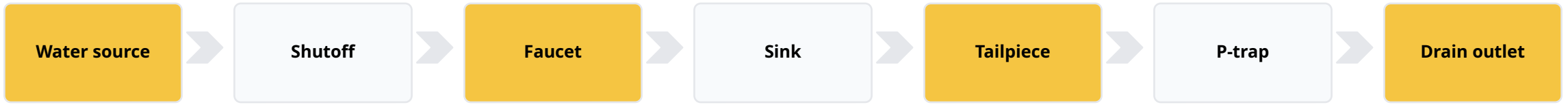
The cut is only as good as the layout.

Read the tape correctly.
Mark the waste side.
Use the square for straight lines.
Verify with a partner before cutting.
Write important dimensions down.

Mark it so someone else can understand it.

Plumbing system example

Today's lab uses plumbing because leaks tell the truth quickly.



Supply brings clean water in. Drainage carries used water out. The trap protects the space.

Why a P-trap exists

The bend is not decorative. It is a simple safety system.

Holds water to block sewer gas.
Creates a removable service point.
Needs correct washer orientation and tight joints.
Leaks usually come from alignment, missing washers, or overtightening.

The part has a job. Know the job before tightening.

Afternoon lab mission

Rolling sink module: assemble, test, correct, document.

1. **Inventory parts**
2. **Dry-fit layout**
3. **Assemble by hand**
4. **Leak test**
5. **Troubleshoot**
6. **Document**
7. **Reset station**

Build - test - correct - document.

Lab roles

Clear roles prevent six hands from becoming one plumbing octopus.

Builder

Assembles parts and calls out each step.

Inspector

Checks alignment, washers, tightness, and leaks.

Documenter

Records parts, problems, fixes, and evidence photos.

Common mistakes

Mistakes are data when they are noticed early.

- Cross-threaded slip nut
- Washer missing or reversed
- Trap arm not aligned
- Overtightened plastic fitting
- No bucket or towel during testing
- Leak not documented before repair

Find it early. Fix it clean.

Verification checkpoint

Stop before water. Inspect before testing.

All washers present and facing correctly.
Slip nuts started straight.
Trap path can be explained.
Bucket/towel ready.
Instructor or peer check complete.

No water until the path makes sense.

Verification log

Documentation is safety support, not paperwork theater.

What I checked
What I expected
What I observed
Tool or method used
What I corrected
What still needs checking

If it is not documented, the next person has to guess.

Cleanup standard

The job is not done until the next person can start safely.

Water off and area dry
Parts counted and returned
Tools cleaned and stored
Floor clear of trip/slip hazards
Evidence uploaded or turned in

Reset is part of the work.

Exit ticket

Write complete sentences. Short is fine. Vague is not.

One residential system I can name is…
One safety rule I need to remember is…
One thing I verified during lab was…
One thing I want more practice with is…

This tells us what to reteach.

Tomorrow preview

Day 2: measuring, layout, framing, and controlled tool use.

Tape measure accuracy
Speed square layout
Cut list basics
Wall parts: plates, studs, rough opening
Power tool clearance and supervised cuts
Building toward the individual wall project

Tomorrow starts the wall.